

Manush Prajwal

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PROFESSIONAL SUMMARY

Software engineer passionate about building scalable backend systems, cloud infrastructure, and AI-powered applications using Node.js, NestJS, PostgreSQL, Docker, and AWS. Founding Engineer of **Atlas Bill**, a multi-tenant retail ERP serving **26 stores**, generating **INR 104K in revenue within 47 days**, and processing over **INR 600K in monthly transaction value**. **4X Hackathon Winner** and author of **11 research publications**, 9 IEEE conference papers and 2 journal papers.

SKILLS

Backend & Systems	Node.js, NestJS, REST APIs
Cloud & DevOps	AWS (EC2, RDS, S3, Lambda, CloudWatch), GCP, Docker, Terraform, Prometheus
AI & Agentic Systems	LangChain, LangGraph, PhiData, LangFlow, LLM Orchestration, n8n Workflows
Databases	PostgreSQL, Supabase, MongoDB
Languages	Python, TypeScript, JavaScript, SQL, C, C++
Developer Tools	Git, GitHub, Linux, Postman

EXPERIENCE

Research Intern — Machine Learning & Photonic Biosensors Oct 2023 – Jan 2024
IEEE Photonics Society, Bengaluru

- Achieved **90.03% diabetes-detection accuracy** by engineering a diagnostic framework that integrated machine-learning models with photonic biosensor measurements.
- Improved biosensor data quality by reducing signal noise by **40%** through preprocessing, normalization, feature extraction, and experimental validation pipelines.
- Presented validated research outcomes at **INDIACom 2024** and **IEEE ICCCNP 2025**.

EDUCATION

The Oxford College of Engineering Sept 2023 – May 2027
B.E. in Computer Science and Engineering (**CGPA: 8.3/10**)
Expected May 2027
Relevant Coursework: Distributed Systems, Database Management Systems, Operating Systems, Machine Learning

Alvas PU College, Moodubidri 2021 – 2023
Pre-University Course in PCMC (**Percentage: 91.6%**)

ACHIEVEMENTS

- 4X Hackathon Winner** — Google, BNMIT, RNSIT, TOCE [Certificates](#) | [IEEE Xplore](#)
- 3rd Runner-Up** — Google GenAI Exchange Hackathon 2025 among **278K+ developers** and **4,457 prototypes** submitted.
- Research Publications** — Author of **11 research papers (9 IEEE conference, 2 journals)** across machine learning, distributed systems, and secure computing architectures.
- Built **Atlas Bill ERP platform**, generating **INR 104K revenue in 47 days** with **26+ stores onboarded**.

PROJECTS

Atlas Bill — Multi-Tenant Retail ERP Platform [Live](#) | [GitHub](#)

- Onboarded **26 supermarkets across Karnataka and Tamil Nadu** and generated **INR 104K in revenue within 47 days** by engineering and launching a production-ready retail ERP platform.
- Reduced manual operational effort by **65%** by automating billing, inventory, CRM, analytics, audit tracking, financial reporting, and Tally workflows.
- Processed over **INR 600K in monthly transaction value** by developing real-time inventory synchronization and automated billing workflows across multiple stores.
- Supported isolated and concurrent operations for **26 stores** by architecting a multi-tenant backend.

AWS CloudPilot — Agentic FinOps & Infrastructure Optimizer [Live](#) | [GitHub](#)

- Built an agentic system that analyzed AWS infrastructure metrics and generated Terraform-based right-sizing recommendations. Identified estimated cloud cost-saving opportunities of up to **30–40% in simulated workloads**.
- Automated cloud monitoring, reasoning, planning, and execution by architecting a four-stage agentic workflow using **CloudWatch, Prometheus, Terraform, and the AWS SDK**.
- Detects idle cloud resources by implementing cost-anomaly detection and resource-utilization analysis across simulated AWS environments.
- Prevented unsafe infrastructure modifications by designing a sandboxed Terraform execution environment with policy-based guardrails, human approval gates, validation checks, and automated rollback.